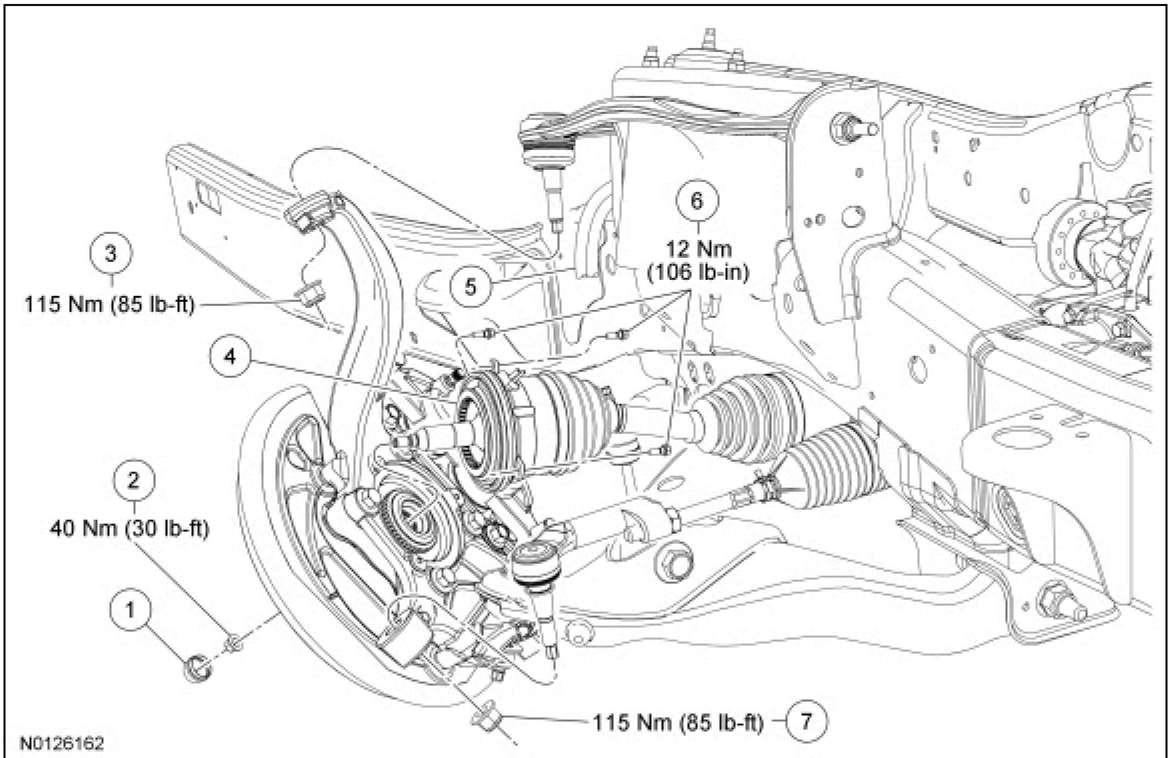
Integrated Wheel End (IWE)

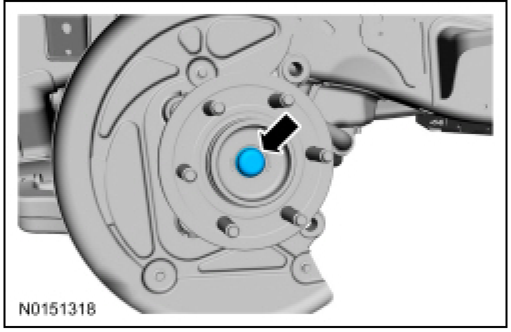
Ford Ecat Electronic Parts Catalog



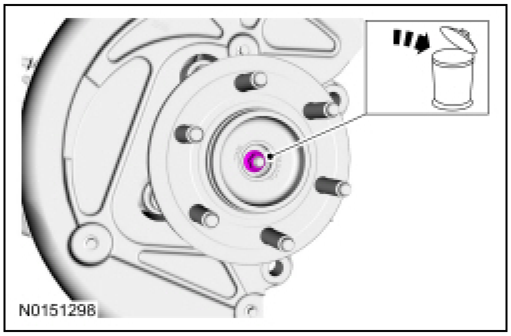
Item	Part Number	Description
1	113111131	Dust cap
2	N802827N802827-S-S	Axle nut
3	W520215W520215-S-S	Upper ball joint nut
4	3C2473C247	Integrated Wheel End (IWE)
5	—	Vacuum vent and supply tube
6	W710279W710279-S-S	IWE bolts (3 required)
7	W520215W520215-S-S	Tie-rod nut

Removal

1. Remove the brake disc. Refer to Section 206-03.
2. Remove the dust cap.



3. Remove and discard the axle nut.

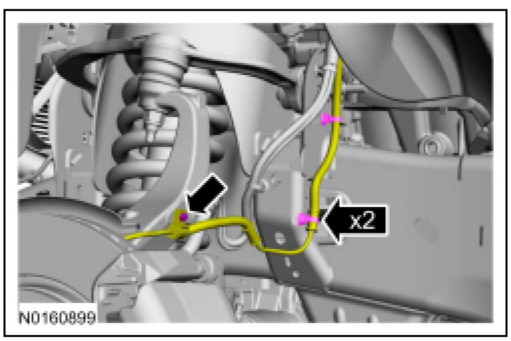


4. Disconnect the vacuum tubes from the Integrated Wheel End (IWE) .

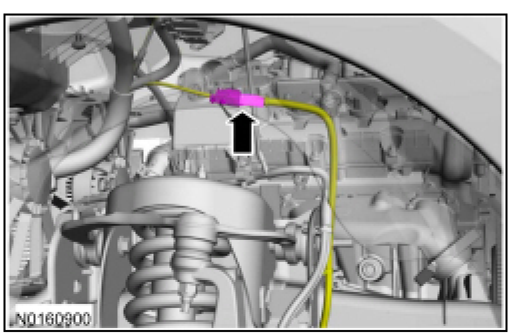
5. Remove the 3 IWE -to-wheel knuckle retaining bolts.

6. Remove the tie-rod nut and separate the tie rod from the wheel knuckle.
   - Discard the tie-rod nut.

7. Remove the wheel speed sensor harness bracket bolt. Detach the wheel speed sensor retainers.



8. Disconnect the wheel speed sensor harness and position the harness aside.



9. Remove the front brake flexible hose bracket bolt and position the brake hose aside.

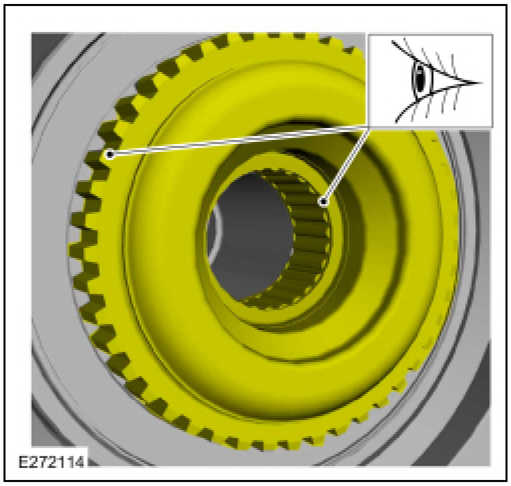
10. Remove the upper ball joint nut and separate the upper ball joint from the wheel knuckle.
    - Discard the upper ball joint nut.

11. **NOTE:** *Allow the wheel knuckle to swing outward while keeping the halfshaft pushed inward.*

Once clearance is available, remove the halfshaft outboard end from the wheel knuckle hub bearing.

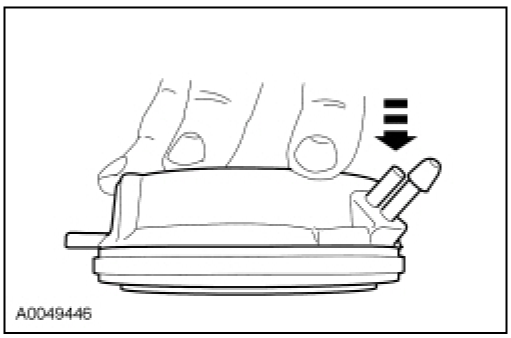
12. Remove the IWE from the halfshaft outboard end.

13. Inspect the hub engagement teeth and the wheel bearing for wear or damage.



Installation

1. Compress the IWE and install a vacuum cap on the vacuum port.



2. **NOTICE: Do not dislodge the Integrated Wheel End (IWE) seal spring when installing the IWE on halfshaft outboard end or component damage may occur.**

   Install the IWE onto the halfshaft outboard end making sure the Integrated Wheel End (IWE) splines line up to the outboard halfshaft splines.

3. **NOTE:** *Allow the wheel knuckle to swing outward while keeping the halfshaft pushed inward.*

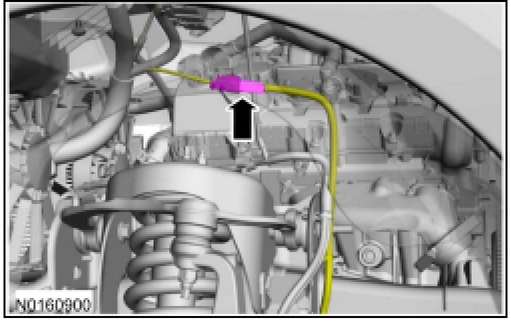
   Once clearance is available, install the halfshaft outboard end into the wheel knuckle hub bearing.

4. Connect the upper ball joint and install a new nut.
   - Tighten to 115 Nm (85 lb-ft).

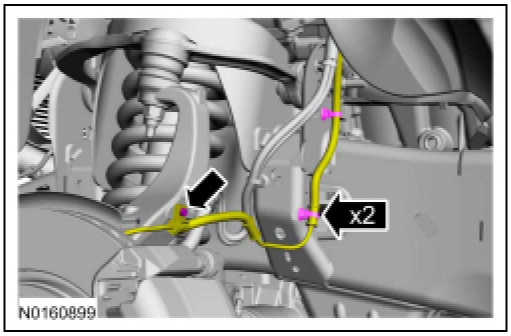
5. Connect the tie rod and install a new nut.
   - Tighten to 115 Nm (85 lb-ft).

6. Position the front brake flexible hose bracket and install the bolt.
   - Tighten to 12 Nm (106 lb-in).

7. Position the harness and connect the wheel speed sensor harness.



8. Attach the wheel speed sensor retainers and install the wheel speed sensor harness bracket bolt.
    - To install, tighten to 12 Nm (106 lb-in).

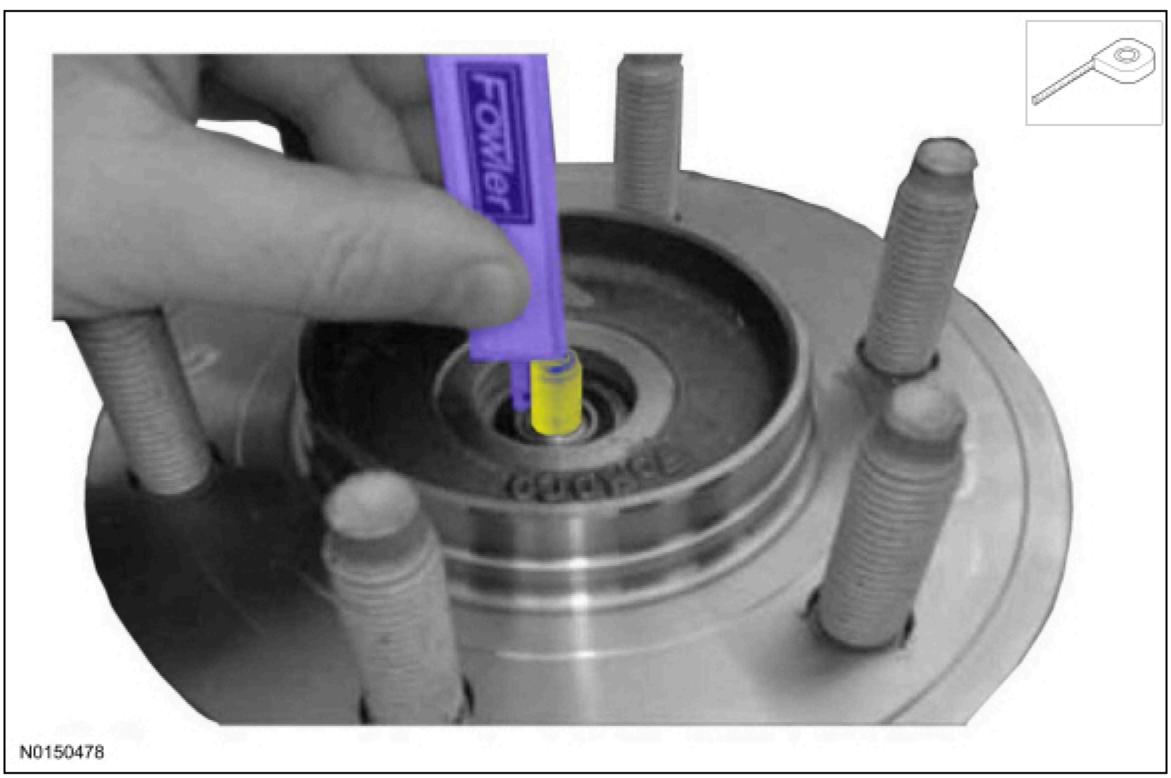


9. Install the 3 IWE -to-wheel knuckle retaining bolts.
    - Tighten to 12 Nm (106 lb-in).

10. Remove the IWE vacuum cap and connect the vacuum tubes.

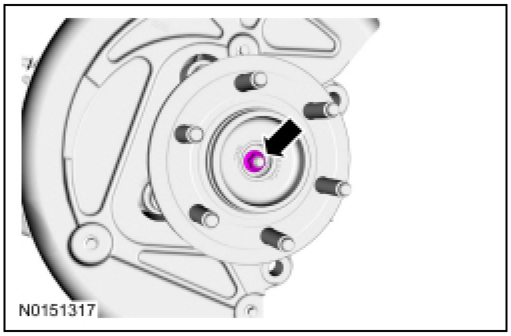
11. **NOTICE: Measure the depth of the CV Shaft threaded end to the inner bearing race (shown in illustration). The minimum depth is 15.5 mm (0.61 in). If the depth is less than 15.5 mm (0.61 in) rotate the CV shaft to clear a binding condition between the IWE and CV splines. Installing the axle nut and tightening without the proper depth of protrusion will result in damage to the IWE .**

    Measure the CV shaft threaded end to the inner bearing race.

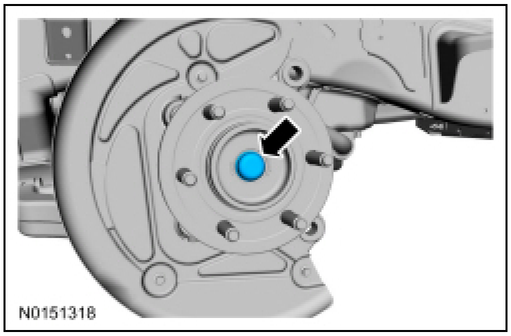


12. **NOTICE: Verify the spline engagement by checking for spline lash before installing the axle nut or component damage may occur.**

Install the new axle nut.
- Tighten to 40 Nm (30 lb-ft).



13. Install the dust cap.



14. Install the brake disc. Refer to Section 206-03.